

13 Weeks to Working PoC. Week 7: Live. Week 13: First Revenue.

Week-by-week implementation plan incorporating all P0 and P1 fixes from the v2 gap analysis: project persistence, real formation presets, LAS file import, geomechanical module, jurisdiction toggle, onboarding tutorial, and provisional patent filing.

13-Week PoC Sprint Plan

BUILD ORDER PRINCIPLE

Weeks 1–2: Data & classical model foundations (Python, offline). **Weeks 3–7:** Visual product — Three.js studio, presets, save/load, deploy. **Weeks 8–10:** ML extraction, LAS import, geomechanical module. **Weeks 11–13:** Jurisdiction toggle, onboarding, outreach, first revenue. Visual experience ships at Week 7 with placeholder physics. ML wires in Week 10. Patent files at Week 7 before demo goes public.

WEEK	PRIMARY FOCUS	KEY DELIVERABLE	PRIORITY
1	IFT data extraction + Python ML environment	Cleaned dataset CSV (800+ rows). MLflow configured.	Foundation
2	Classical model implementations (Span-Wagner, Duan-Sun, Li et al.)	3 Python baseline models validated against CoolProp/NIST	Foundation
3	Vite + React + Three.js scaffold + Zustand store + JSON save/load	Running dev server. Export/import project JSON working. P0 ✓	P0
4	Three.js reservoir renderer + formation presets (Sleipner, Decatur, Otway)	3D formation renders. One-click presets load all parameters. P0 ✓	P0
5	Parametric geometry (anticline, dome, layered) + well placement raycaster	Users select geometry type, click to place wells	Core
6	Web Worker simulation engine (placeholder physics) + D3.js 2D toggle	CO ₂ plume animating from well. 3D ↔ 2D switch working.	Core
7	Cloudflare KV cloud save + onboarding tutorial overlay + deploy + patent	carbonlens.io LIVE. Cloud save working. Provisional patent filed. P0 ✓ IP ✓	P0 + IP
8	Train GMDH/MGGP/MARS on IFT dataset. Extract equations to TypeScript.	predictIFT.ts running in browser, validated vs Python (<0.5% dev)	ML
9	Extract all 7 property equations. Wire to simulation engine. Cloudflare Worker for ensemble weights (hybrid architecture).	All 7 properties live from ML. Ensemble weights protected server-side.	ML + IP
10	LAS 2.0 parser + geomechanical tab (FPG, Eaton, MAIP, seal index)	LAS upload auto-populates formation. MAIP computed from Eaton. P1 ✓	P1
11	Jurisdiction toggle (EPA/UK NSTA/NPD/PETRONAS) + permit export	4 jurisdiction output templates. PDF export. P1 ✓	P1
12	Stripe billing (Free/Researcher/Professional gating) + P10/P50/P90 envelopes wired to Three.js	Payment gating live. Uncertainty envelopes render on plume.	Revenue

13	30 cold emails (EPA tracker). Submit JPSE paper. SPE abstract submitted.	First trial signups. Revenue pipeline started. Academic citation in pipeline.	GTM
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Foundation → Live Deployment

Week 1 Data Collection & Python ML Environment

7 hrs

- ☐ Extract IFT data tables from Papers A1–A5 using tabula-py. Target: 800+ rows across T/P/salinity space. [DATA](#)
- ☐ Set up Python ML environment: scikit-learn, pyearth, gmdh, geppy, CoolProp, MLflow. Configure local MLflow tracking server. [CODE](#)
- ☐ Generate 10K-point Span-Wagner density grid using CoolProp across T: 0–150°C, P: 1–100 MPa. Save as CSV for ML training. [DATA](#)
- ☐ Read all 9 EPA Class VI guidance documents. Extract the specific calculation requirements for MAIP, AoR, and MVA plan sections. [READ](#)

Week 2 Classical Model Implementations & Baselines

7 hrs

- ☐ Implement Span-Wagner EOS in Python. Validate against CoolProp within 0.1% across full T/P range. Log in MLflow. [CODE](#)
- ☐ Implement Duan-Sun solubility model. Validate against NIST Solubility Database within 2%. [CODE](#)
- ☐ Implement Li et al. IFT classical correlation. Establish $R^2=0.94$ baseline on IFT dataset. [CODE](#)
- ☐ Implement Eaton geomechanical correlation in Python. Validate against published Sleipner geomechanical data. [CODE](#)

Week 3 – P0 Vite Scaffold + Zustand Store + Project Save/Load

7 hrs

- ☐ **P0** Scaffold: `npm create vite@latest carbonlens-studio -- --template react-ts`. Install Three.js, D3.js, Zustand, Tailwind. Confirm dev server running. [CODE](#)
- ☐ **P0** Define Zustand store schema covering all reservoir parameters, well positions, simulation state, UI view mode, jurisdiction selection, project metadata. [CODE](#)
- ☐ **P0** Implement JSON export: “Save Project” button serialises full Zustand store to .carbonlens JSON file. Download via browser. “Load Project” reads file, hydrates store. Test round-trip. [CODE](#)
- ☐ Build parameter input panel (Tailwind): all formation inputs with units, validation, and help tooltips referencing source papers. [CODE](#)

Week 4 – P0 Three.js Renderer + Real Formation Presets

7 hrs

- ☐ Build Three.js reservoir renderer: BoxGeometry with layered MeshPhongMaterial, OrbitControls, directional lighting. Formation renders live from Zustand parameter changes. [CODE](#)
- ☐ **P0** Implement Sleipner preset JSON with all Utsira Sand parameters, published P50 plume extent at 10/20/50 years, citation metadata. One-click load populates all inputs and renders. [CODE](#)
- ☐ **P0** Implement Decatur (Illinois Basin) and Otway Basin presets with same structure. All three presets accessible from “Formation Library” panel. [CODE](#)

- ☐ Add OrthographicCamera for 2D view mode. Toggle 3D/2D seamlessly from view selector. [CODE](#)

Week 5 Parametric Geometry + Well Placement

7 hrs

- ☐ Build anticline geometry: sine-wave surface deformation applied to BoxGeometry vertex positions. Dip angle and amplitude controlled from parameter panel. [CODE](#)
- ☐ Build dome geometry (radial sine) and layered aquifer (flat strata with alternating permeability layers). All three geometry types selectable. [CODE](#)
- ☐ Implement well placement: Three.js Raycaster on canvas click. Click on formation surface places CylinderGeometry well with depth, rate, and perforation interval inputs. Support up to 5 wells in PoC. [CODE](#)

Week 6 Web Worker Simulation + D3.js 2D Cross-Section

7 hrs

- ☐ Build Web Worker simulation engine: 50x50 grid, simplified Darcy flow with hardcoded placeholder CO₂ density and viscosity values. Posts saturation arrays to main thread each tick. [CODE](#)
- ☐ Wire saturation arrays to Three.js vertex colour updates: 0% saturation = formation colour, 100% = red, interpolated via Viridis colourmap. Plume visually advances from well. [CODE](#)
- ☐ Build D3.js 2D vertical cross-section view: colour-mapped SVG grid, formation layers, caprock line, well symbol, plume extent. Toggle from Three.js view. [CODE](#)
- ☐ Add playback controls: Play/Pause/Reset simulation, timestep slider, years-elapsed counter. [CODE](#)

Week 7 – DEPLOY + IP Cloud Save + Onboarding + Deploy + Patent Filing

7 hrs

- ☐ **PO** Implement Cloudflare KV cloud save: Cloudflare Worker receives POST with Zustand JSON + Stripe customer ID, stores in KV. Load on login from any device. [CODE](#)
- ☐ Build onboarding tutorial overlay: 6-step guided sequence for first-time users — loads Sleipner preset, explains each parameter, places a well, runs simulation, shows results. Skip option for returning users. [CODE](#)
- ☐ Register carbonlens.io (or carbonlens.app). Deploy Vite build to Cloudflare Pages. Configure custom domain. Confirm CDN delivery from Malaysia, UK, US, Norway. [BIZ](#)
- ☐ **IP** File USPTO provisional patent application. Claims: (1) GMDH+MGPP+MARS ensemble with bootstrap P10/P50/P90, (2) hybrid client-side/server-side architecture for equation IP protection, (3) browser-based CO₂ plume simulation driven by white-box ML properties. \$320 micro-entity fee. [BIZ](#)

✓ Week 7 Milestone: carbonlens.io is LIVE

- ▲ Working 3D reservoir builder. Sleipner/Decatur/Otway presets. Project save/load. Guided onboarding. Provisional patent filed. Share demo link with 5 CCUS engineer contacts for immediate feedback.

ML Wiring → First Revenue

Week 8

Train ML Models + TypeScript Equation Extraction

7 hrs

- ☐ Train GMDH, MGGP, MARS on IFT dataset. Target $R^2 > 0.98$ for ensemble. Log all experiments in MLflow. [CODE](#)
- ☐ Extract MGGP symbolic expression to TypeScript function: `const predictIFT_mggp = (T: number, P: number, S: number): number => ...`. Print equation from geppy, transcribe exactly. [CODE](#)
- ☐ Extract MARS basis functions: export as array of hinge function objects, implement in TypeScript as computed if-else logic. [CODE](#)
- ☐ Validate: run TypeScript IFT prediction vs Python model on full test set. Confirm $<0.5\%$ deviation at every validation point. [CODE](#)

Week 9

Hybrid Architecture + All 7 Properties Live

7 hrs

- ☐ Implement hybrid IP architecture: classical baseline equations (Span-Wagner, Duan-Sun, Fenghour, Garcia, Li et al.) in TypeScript client-side. Ensemble P10/P50/P90 weights deployed as Cloudflare Worker endpoint. [CODE](#)
- ☐ Wire all 7 properties to simulation Web Worker: replace hardcoded placeholder values with live ML-corrected outputs. Confirm simulation physics update correctly as formation params change. [CODE](#)
- ☐ Build property display panel: IBM Plex Mono readouts for all 7 properties, updating live. Each value shows P50 with $\pm$ uncertainty range. Source citation badge per property. [CODE](#)
- ☐ Add P10/P50/P90 plume envelope rendering: three overlaid colour contours in Three.js (blue = P10 conservative, teal = P50, amber = P90 worst-case). [CODE](#)

Week 10 – P1 LAS 2.0 Import + Geomechanical Module

7 hrs

- ☐ **P1** Build LAS 2.0 file parser in TypeScript: reads header section (metadata, curve units), data section (depth-indexed values). Extracts TVD, PHIE, RHOB, temperature from geothermal gradient. Auto-populates formation panel. [CODE](#)
- ☐ Implement curve-to-parameter mapping UI: show parsed curves, allow user to confirm or reassign each curve to the correct formation parameter. Handle unrecognised curves gracefully. [CODE](#)
- ☐ **P1** Build geomechanical tab: overburden stress (σ_v), Eaton minimum horizontal stress (σ_h), FPG calculation, MAIP (FPG $\times$ depth $\times$ 0.9 safety factor). All computed client-side. [CODE](#)
- ☐ Wire caprock seal index to IFT Model 1 output: CO_2 column height = $(2\gamma\cos\theta)/(rpg)$. Updates live as IFT changes with T/P/salinity. Display in geomechanical tab alongside MAIP. [CODE](#)

Week 11 – P1 Jurisdiction Toggle + Permit Export

7 hrs

- ☐ **P1** Build jurisdiction dropdown: EPA Class VI / UK NSTA / Norwegian NPD / PETRONAS Malaysia. Changing selection updates: output template structure, nomenclature labels, pressure unit display (MPa vs psi), depth reference. [CODE](#)

- ☐ Implement permit export: MAIP, AoR radius at P10/P50/P90, CO₂ column height, phase state, property table with citations. Format matches selected jurisdiction template. Export to PDF via pdfmake (client-side). [CODE](#)
- ☐ Add PETRONAS Malaysia preset formation (Offshore Sarawak Basin parameters from published PETRONAS CCS feasibility studies). List as “beta” preset. [DATA](#)
- ☐ Deploy updated build to Cloudflare Pages. Share updated demo link with 10 CCUS engineering contacts. [BIZ](#)

Week 12 Stripe Billing + Uncertainty Envelopes Complete

7 hrs

- ☐ Implement Stripe billing: Free (Explorer) / Researcher \$99 / Professional \$499 tier gating via Cloudflare Worker JWT validation. Stripe Checkout for upgrade flow. [BIZ](#)
- ☐ Gate features by tier: 3D simulation → Professional+; permit export → Researcher+; LAS import → Professional+; cloud save → Researcher+. Free tier gets 2D + presets + JSON export only. [CODE](#)
- ☐ Implement uncertainty sigma slider in UI. Wire to P10/P50/P90 envelope width in both Three.js 3D and D3.js 2D cross-section. Add Monte Carlo histogram in sidebar (n=200, runs in Web Worker). [CODE](#)
- ☐ Polish: loading states, error boundaries, mobile responsiveness check, keyboard shortcuts for Play/Pause/Reset. [CODE](#)

Week 13 Outreach + Academic Citation + Revenue Pipeline

7 hrs

- ☐ Send 30 personalised cold emails to EPA Class VI permit applicants identified from EPA public tracker. Include direct link to Decatur preset demo. Subject: “Free CO₂ storage simulation tool for your [Project Name] application.” [BIZ](#)
- ☐ Email 3 UTP CCUS faculty + 2 international collaborators (University of Wyoming, UT Austin). Offer free Studio access in exchange for loading the Sleipner preset and providing written feedback for a co-authored validation paper. [BIZ](#)
- ☐ Submit paper to JPSE or IJGGC citing CarbonLens Storage Studio in the methods section as the simulation tool used for property prediction. [WRITE](#)
- ☐ Submit SPE Annual Technical Conference abstract: “CarbonLens Storage Studio: A Browser-Based ML-Powered CO₂ Storage Simulation Platform for Class VI Permit Applications.” [WRITE](#)
- ☐ Post 60-second LinkedIn screen recording: Sleipner preset loading, CO₂ plume simulating in 3D, P10/P50/P90 envelopes, switching to Norwegian NPD jurisdiction output. Target: 5,000+ impressions in CCUS engineering community. [BIZ](#)

WEEK 13 TARGET STATE

Working 3D reservoir studio live at carbonlens.io. All P0 and P1 features shipped. Provisional patent filed. At least 3 trial signups. Revenue pipeline started with at least 2 personalised follow-ups in progress. Academic citation in submission pipeline. The PoC thesis is ready to validate.